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College of Osteopathic  
Medicine | Arkansas



## Pediatric Orthopedics

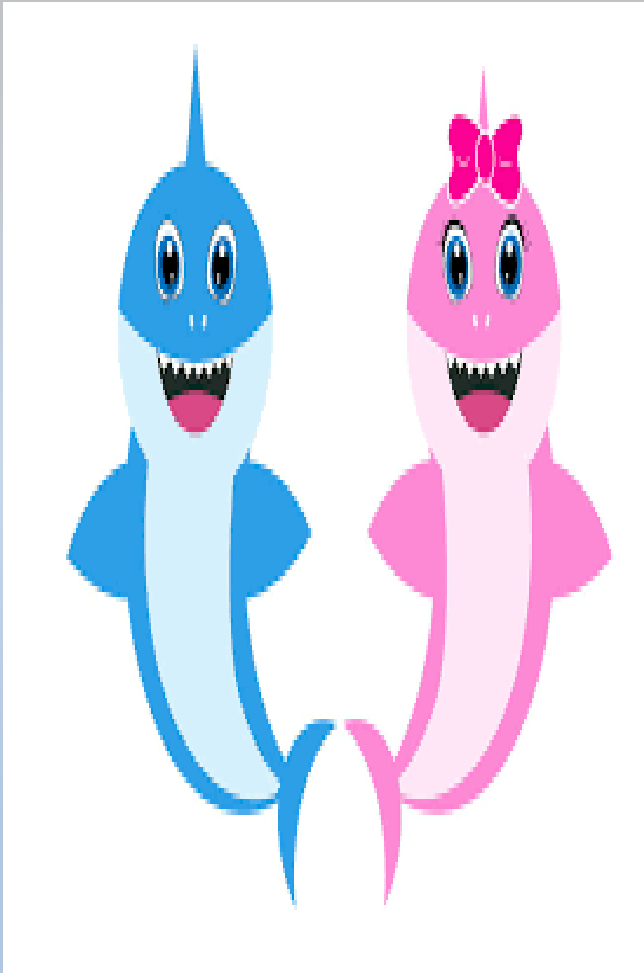
### Principles and Practices of Osteopathic Medicine I

Mitzi Scotten M.D. Associate  
Professor – Arkansas  
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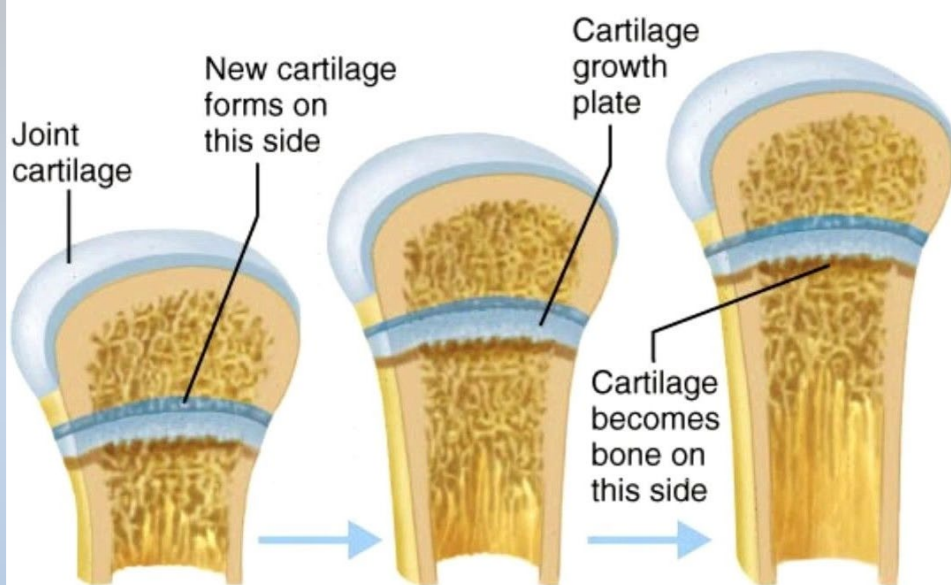
# Session Objectives

1. Discuss the differences in pediatric bone structure and how this affects disease processes
2. Discuss bone fractures in children and how to use the Salter Harris guide for fractures involving the growth plate
3. Discuss the most common developmental and acquired deformities of foot and hip how to diagnose based on age and presentation
4. Review presentation, cause, diagnosis and treatment of pediatric bone and joint infections
5. Review other osteopathic disorders- rickets and Transient Synovitis
6. Identify signs of non-accidental trauma

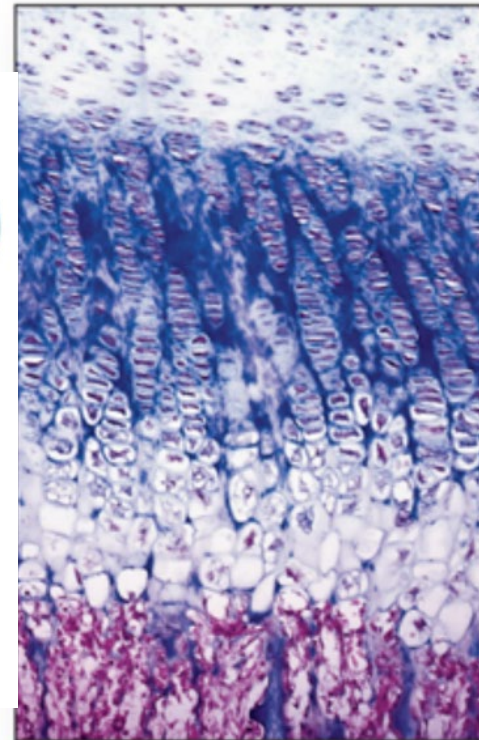
# Baby Shark? ....Not so far off from truth



# Epiphyseal Plate



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.



b Epiphyseal plate

Zone of reserve cartilage

Zone of proliferation

Zone of hypertrophy

Zone of calcified cartilage

Zone of ossification

Source: Anthony L. Mescher: Junqueira's Basic Histology, 14th Edition.  
www.accessmedicine.com  
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# Epiphyseal Plate vs Line



[https://images.radiopaedia.org/images/19224817/194cfaabe33be22d4a5819ac144ed6\\_thumb.jpeg](https://images.radiopaedia.org/images/19224817/194cfaabe33be22d4a5819ac144ed6_thumb.jpeg)

<https://radiopaedia.org/cases/normal> -wrist -paediatric

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# Pediatric Bone are...

**MORE**

**Porous**

**Pliable**

**Thick at the edges**

**Quick to heal**

**LESS**

**Dense**

**Rigid**

**Bending strength**

**Mineralized**



# Pediatric Bone Injuries

- Common
- Related to brain and motor development
- Location based on age
- Intentional vs. Unintentional



# Challenges

Poor Historians  
Uncooperative  
Plain films  
challenging  
Contralateral views  
usually required





# How to Report Out a Fracture

Which bone?

Where in the bone?

What type of  
break?

Describe the  
fracture line

Additional info

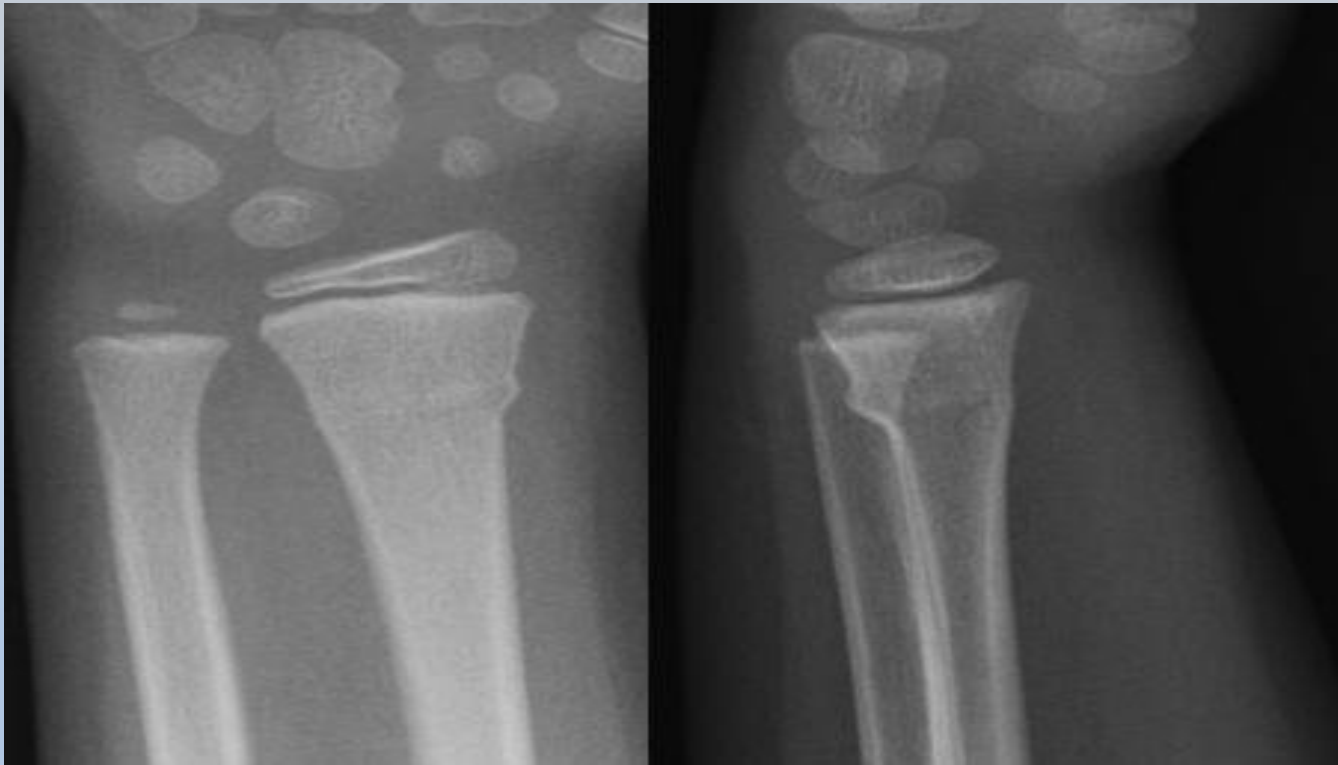




# Common Pediatric Fractures

- **Torus Fracture :**

Common, buckles, compression, incomplete

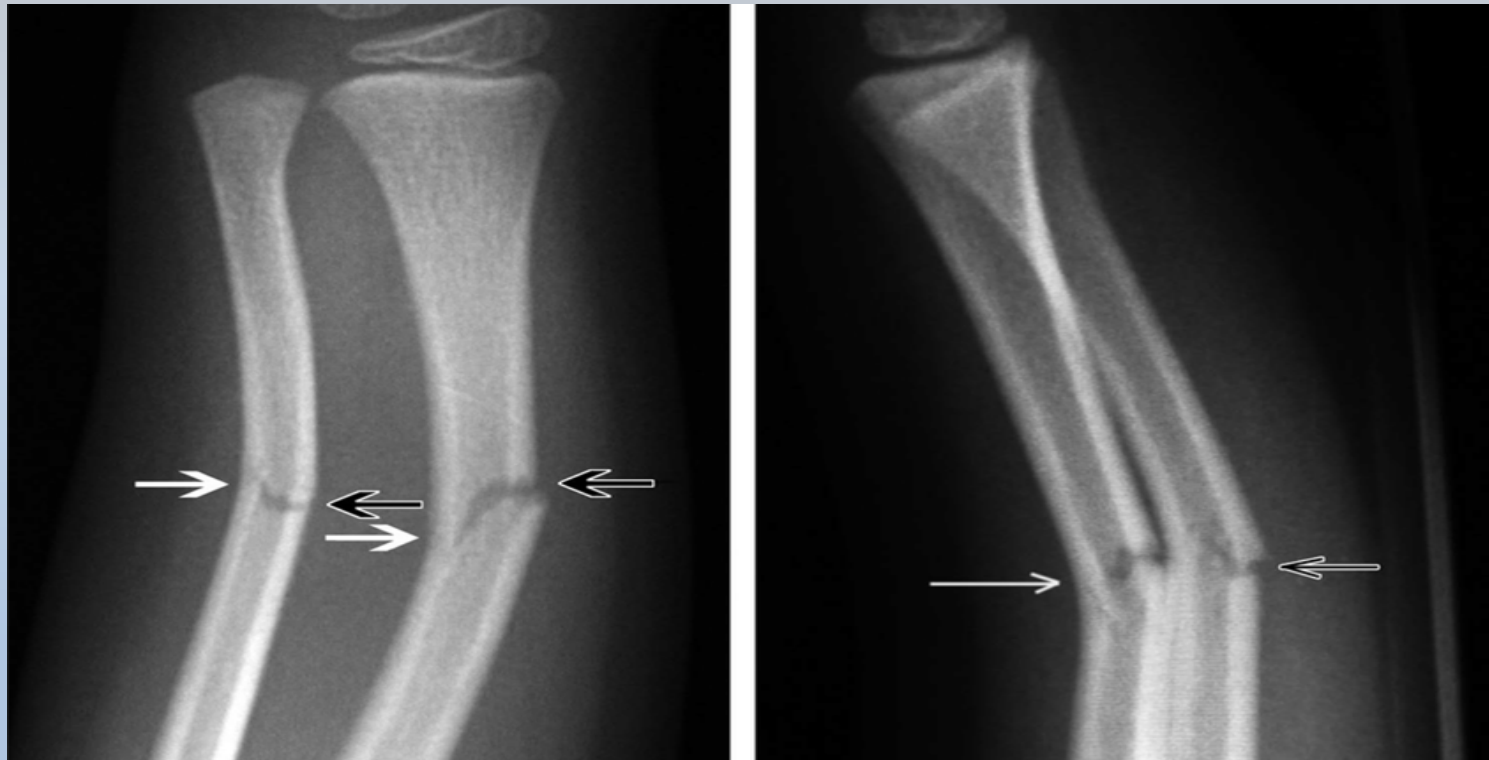


[https://www.researchgate.net/figure/Radiograph-of-torus-fracture-of-distal-radius-anterior-posterior-and-lateral-views\\_fig1\\_325471160](https://www.researchgate.net/figure/Radiograph-of-torus-fracture-of-distal-radius-anterior-posterior-and-lateral-views_fig1_325471160)

# Common Pediatric Fractures

- **Greenstick Fracture:**

Bending, pliant bone, periosteum intact one side



# Salter-Harris Classification of Physeal Fractures



## Injuries Involving the Epiphyseal Plate

BY ROBERT B. SALTER, M.D., F.R.C.S.(C)\*, AND W. ROBERT HARRIS, M.D., F.R.C.S.(C)†,  
TORONTO, ONTARIO, CANADA

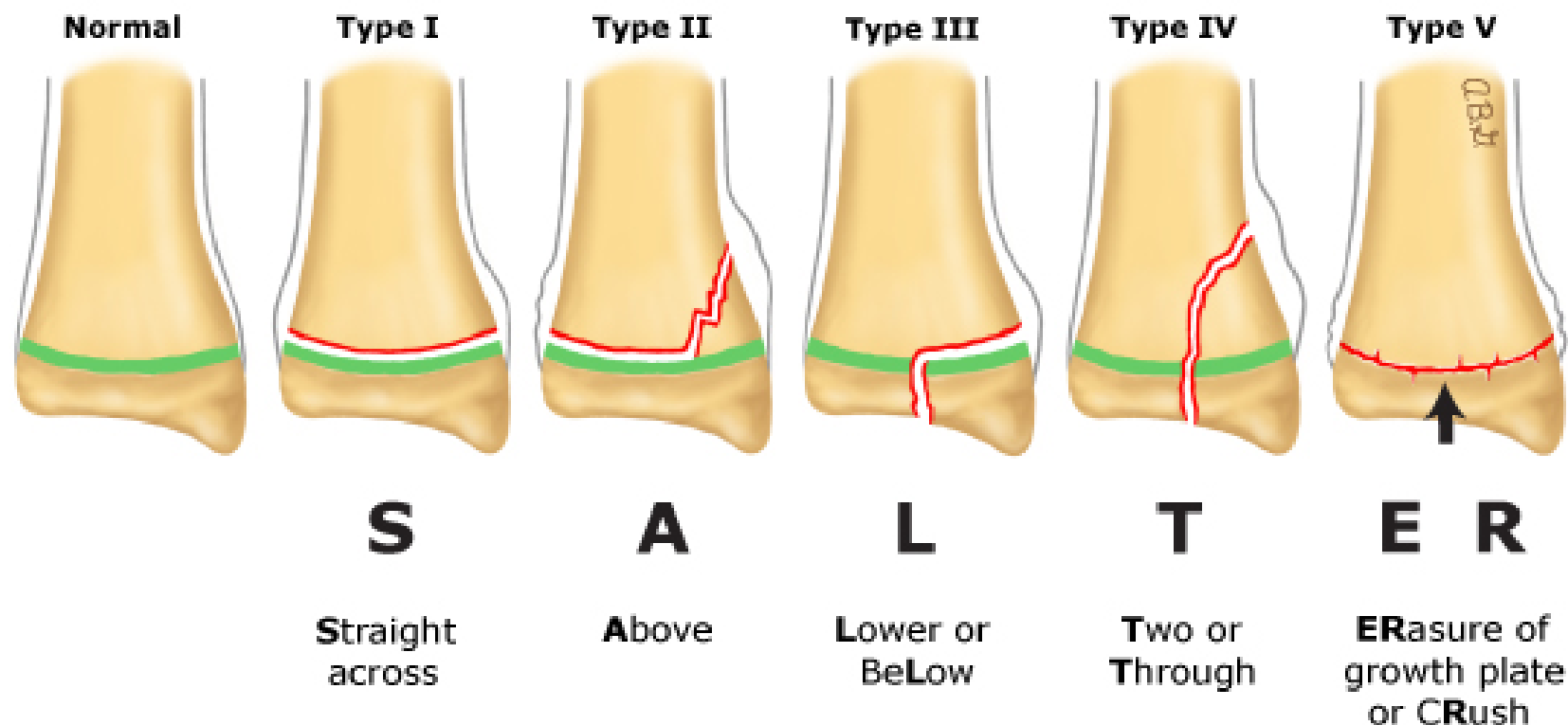
*An Instructional Course Lecture, The American Academy of Orthopaedic Surgeons*

Injuries involving the epiphyseal plate present special problems in diagnosis and management. The dread complication of serious disturbance of growth is usually predictable and, in certain circumstances, can be prevented. Thus, knowledge of the prognosis for a given injury to the epiphyseal plate in a particular child is of considerable importance to the surgeon, who has the dual responsibility of treating the child and advising the parents. The purpose of this presentation is to discuss epiphyseal-plate injuries from both the clinical and the experimental points of view.

*J Bone Joint Surg Am. 1963;45:587-622.*  
ROBERT B. SALTER and W. ROBERT  
HARRIS

# Salter Harris Fractures

## Salter-Harris classification of physeal fractures



- UpToDate, 2018

# Salter Harris Type 1

## Growth plate

- Fracture through the zone of hypertrophy
- Width of the physis increases.
- **Growth disturbances are uncommon**
- **Treat with casting or splinting**



# Salter Harris Type 2

## Metaphysis and growth plate

- **Most common type - 75%**
- Usually will not cause functional limitations.
- Occur above the growth plate



# Salter Harris Type 3

## Epiphysis and growth plate

- Potentially damaging
- Chronic disability due to **extension of fracture into the articular surface**
- Arthritis possible without reduction – surgery
- Treatment surgical.





# Salter Harris Type 4

Epiphysis, growth plate and metaphysis is

- Intra-articular fracture
- May cause chronic pain
- May cause deformity of the joint
- Surgery needed for repair



# Salter Harris Type Five

## Compression or Crush Fracture

- Rare
- Growth affected
- Injury occurs at the growth plate
- X-ray findings are subtle/difficult to see
- High degree of suspicion needed
- Most dx made in retrospect after growth plate sealed





# Non-accidental trauma

**Medical providers are mandated reports of child abuse.**

## Statistics as of 2021

- Over 600,000 children were reported as abused
- Includes mental, physical, sexual and sex trafficking
- Over 70% of abuse is at the hands of a parent
- Youngest are abused the most- 30% under two years
- Children on children abuse is increasing
- Girls generally more vulnerable
- Untreated, abused children go on to be abusers
- Almost 2 thousand children died from their abuse
- Children in abusive home are helpless without teachers, health care workers, friends and relative

# High degree of suspicion

Babies with bruises

Back and torso bruising

Inconsolable crying

Burns

Whip marks

Unkempt and dirty  
appearance

Nervous parent

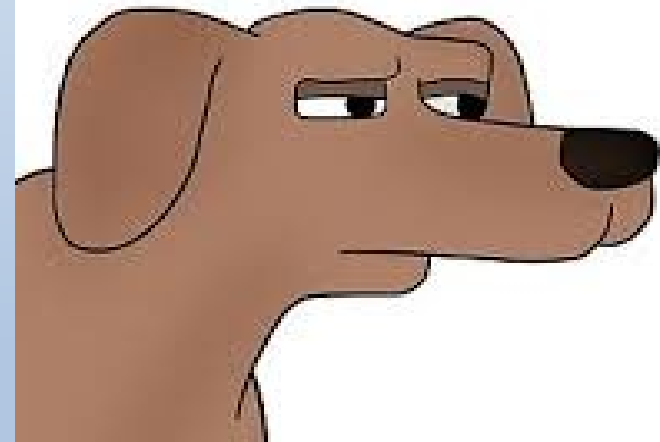
Quiet child with poor eye  
contact

Overly serious demeanor

Parental putting down child

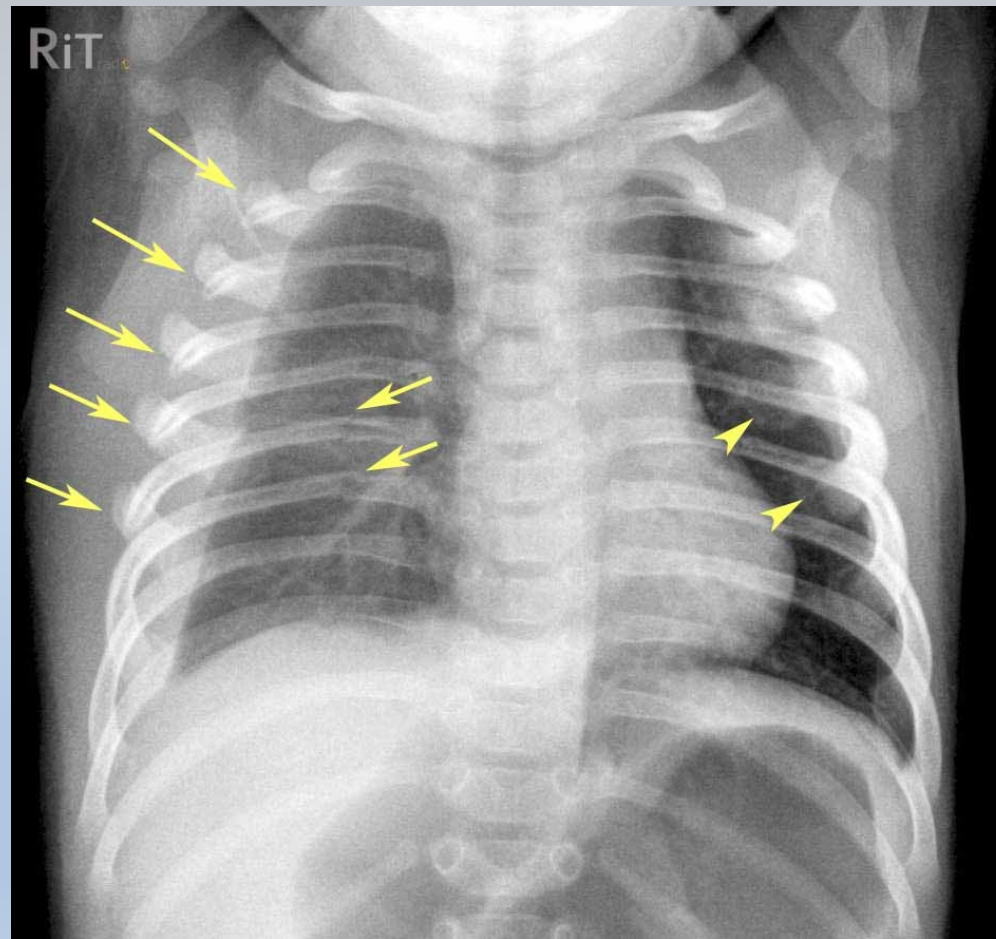
Inappropriate sexual  
behaviors in a young child

**THAT'S SUSPICIOUS...**





# Posterior Rib Fractures



<https://radiologyinthaiblogspot.com/2010/12/non-accidental-injury-nai-rib-fractures.html>

# Skull and Eye Injury

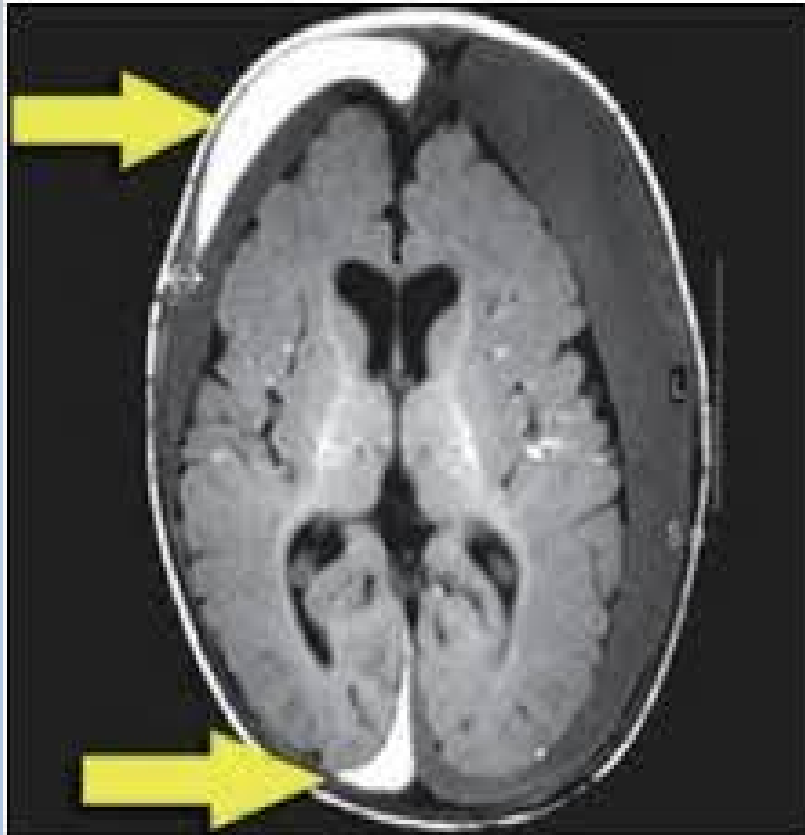
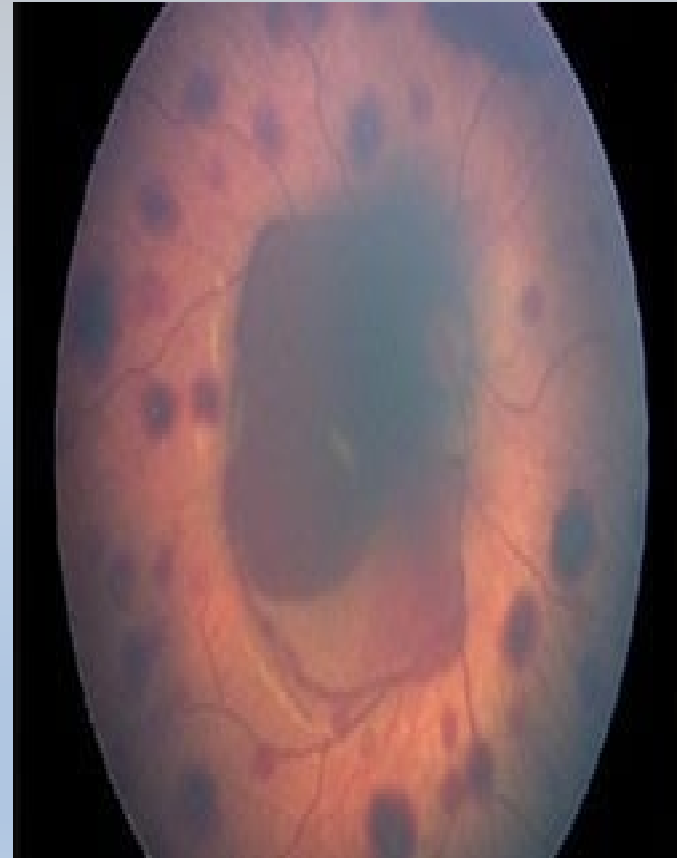


Figure 6 T1-weighted MR image showing bilateral fluid collections from chronic SDH and new SDH in the right frontal and posterior parietal regions.<sup>[8]</sup>



# Long Bone Injuries



[https://www.utmb.edu/pedi\\_ed/Cor eV2/Abuse/page\\_08.htm](https://www.utmb.edu/pedi_ed/Cor eV2/Abuse/page_08.htm)

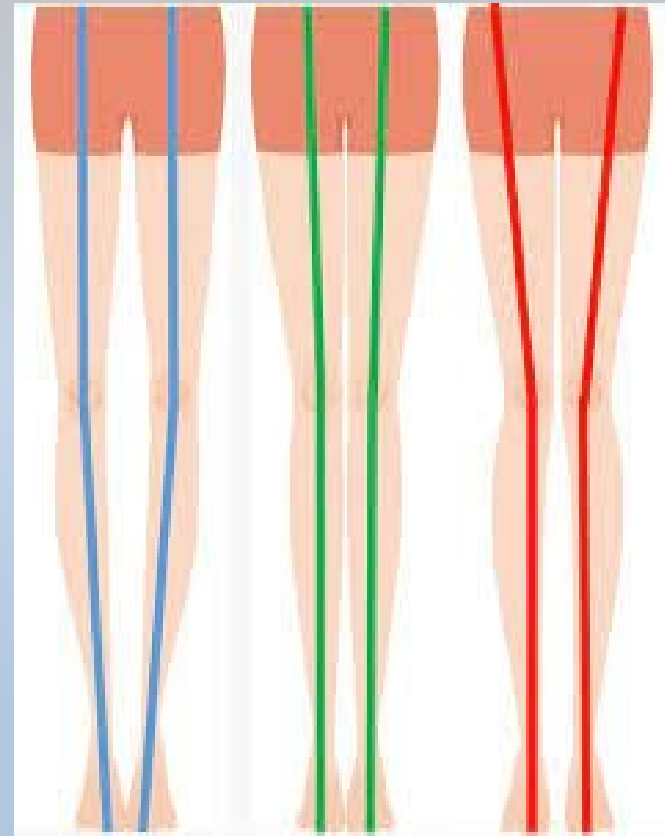


# Orthopedic Disorders

Hip

Knee

Foot



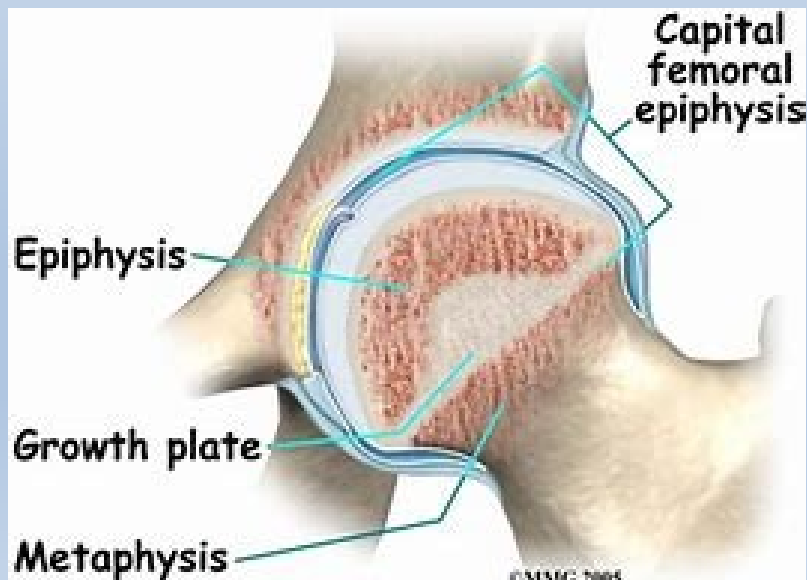


# The Pediatric Hip

## Developmental Dysplasia of the Hip (DDH)

## Legg-Calve-Perthes Disease

## Slipped Capital Femoral Epiphysis (SCFE)



<https://www.rmts.clinic/Injuries-Conditions/Pediatric/Pediatric-Issues/Guide-for-Transient-Synovitis-of-the-Hip-in-Children/a~3061/article.html>

# Developmental Dysplasia of the Hip

## Mechanical instability of the hip joint

Incidence- 1-2/1,000 live births

### Risk Factors:

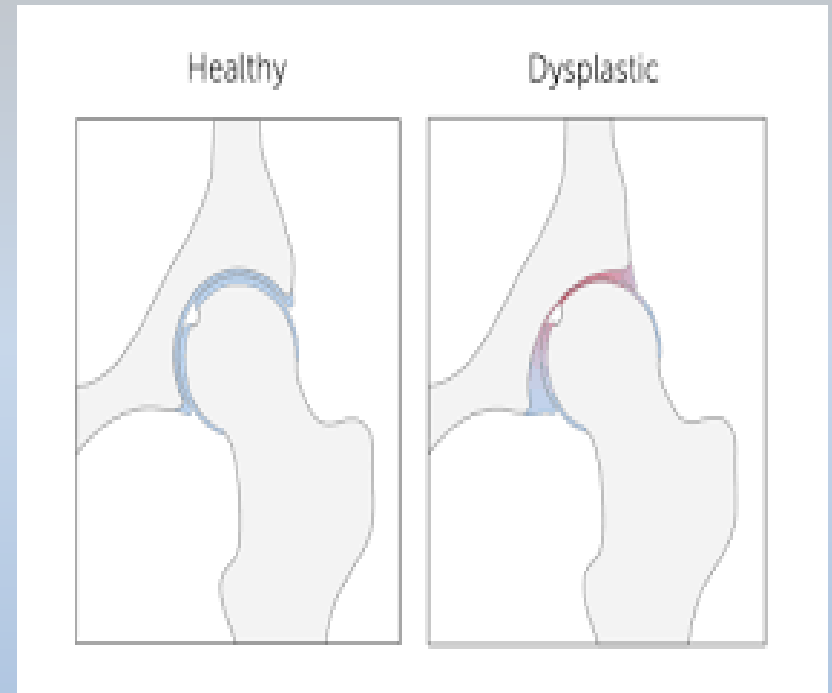
**Female**

**First born**

**Breech**

**Prenatal factors**

**Swaddling**



<https://www.childrenshospital.org/conditions/hip-dysplasia>

# Diagnosing DDH

Diagnosis made with high degree of suspicion

- By special physical exam tests
- By ultrasound

## – Tests to Detect DDH:

- **Barlow Test:** provoke hip prone to dislocation
- **Ortolani Test:** move a dislocated hip back into acetabulum
- **Galeazzi Test:** evaluates for current hip dislocation



Barlow

[musculoskeletalkey.com](http://musculoskeletalkey.com)



Ortolani



# Developmental Dysplasia of the Hip

## Treatment:

- **Pavlik harness** (usually 6-12 weeks)
- \*Ultrasound to verify reduction\*
- **Spica casting**
- Failure of closed reduction->open reduction



<http://www.ossur.co.uk/injury-solutions/products/hip/hip-orthoses/pavlik-harness>

Pavlik Harness



© Healthwise, Incorporated

<https://myhealth.alberta.ca/Health/Pages/conditions.aspx?hwid=ue5665>

Spica Cast

# Legg-Calve-Perthes Disease

- Idiopathic avascular necrosis of femoral head
- Incidence- 1 in 12,000 Male predominant 5X
- Diagnosis made radiographically



# Clinical Presentation

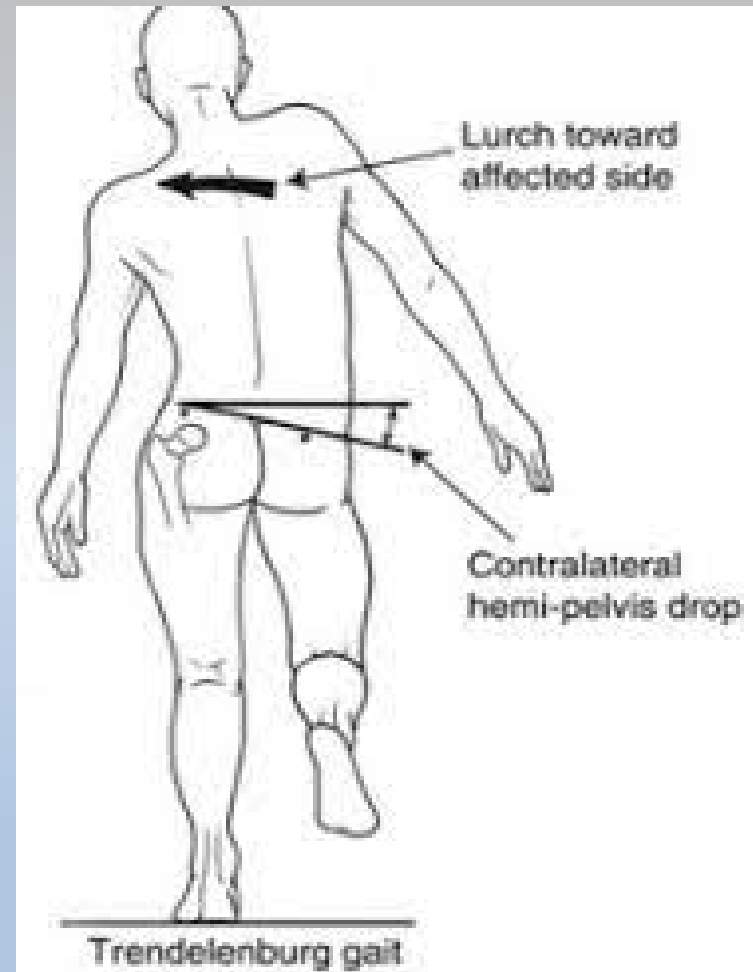
**Pain** (hip, knee, groin,  
and anterior thigh pain)

**Limp** - +/- Pain

**Trendelenburg gait**

**Hip ROM decreased**

Especially with **abduction**  
and **internal rotation**



<https://www.corewalking.com/trendelenburg-gait/>



# Surgery vs. Bracing

**Treatment Goals:** relieve symptoms, restore ROM, and containment of the hip.

- Minimal Weight bearing
- Pain control
- Physical therapy
- Bracing (Petrie Cast) if needed to help contain femoral head within acetabulum
- If severe, surgery will be required



<https://www.pinterest.com/pin/64387469644844752/>

# Slipped Capital Femoral Epiphysis

## Obese, pubertal children during their growth spurt

- Mean age girls: 12
- Mean age boys: 13.5
- ~2x more common in **boys**
- May have positive family history



## Signs & Symptoms:

- Hip, thigh, groin, or knee pain
- Limp
- Externally rotated leg at rest
- **External rotation with flexion**
- Decreased ROM (Flexion and IR)





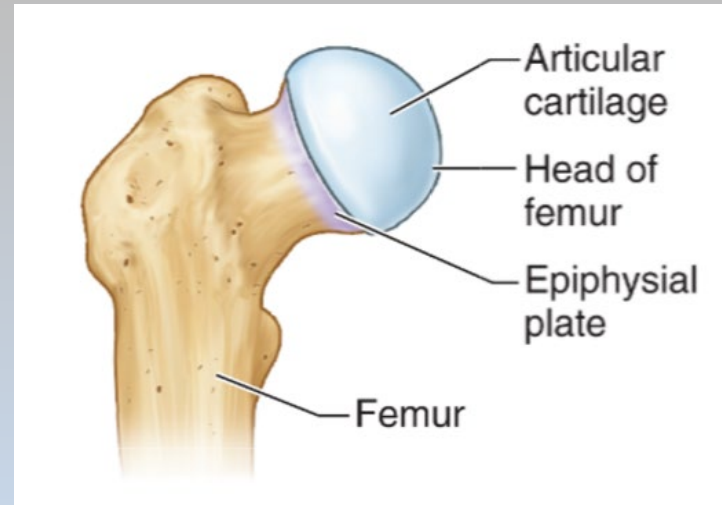
# Slipped Capital Femoral Epiphysis

**Salter Harris type I fracture of the proximal femoral epiphysis.**

Slippage through **zone of hypertrophy** .

Epiphysis stays in the acetabulum.

About 25% of cases can be bilateral.



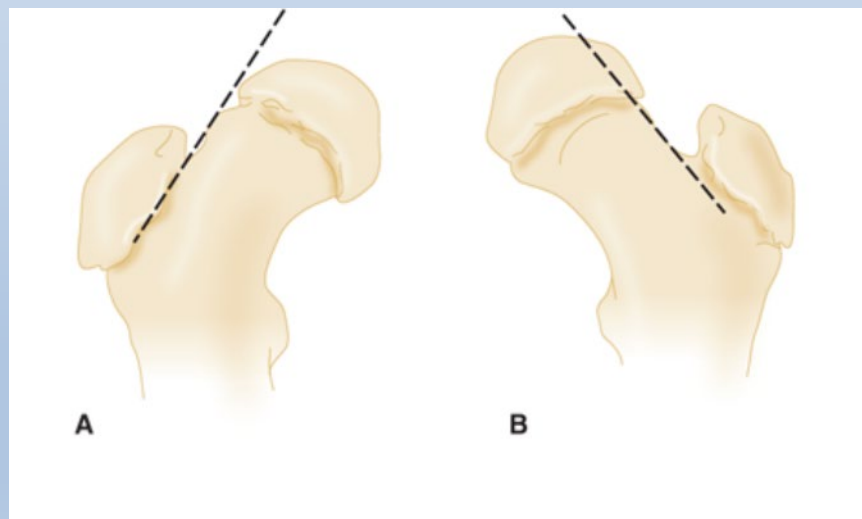
<https://www.reboundca.com/Injuries-Conditions/Pediatric/Pediatric-Issues/Slipped-Capital-Femoral-Epiphysis/a~3380/article.html>

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# Slipped Capital Femoral Epiphysis

- **Diagnosis** via X-ray AP and Frog Leg View
  - AP: **Klein's line** and **Trethowan's sign** (A)



Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline:  
Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition  
[www.accessmedicine.com](http://www.accessmedicine.com)  
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# Treatment

## Surgery is only treatment

Internal fixation using cannulated bone screw(s)

Early treatment prevents morbidity

## Complications:

Avascular necrosis

Early degenerative arthritis



<https://www.wheelessonline.com/joints/treatment-scf/>

# Osgood-Schlatter Disease

Osteochondritis of the **tibial tuberosity**

Common in children ages **9-14 years old**

Pain and swelling at tibial tubercle

**Anterior knee pain** made worse with kneeling, running, and jumping



<https://sportsclinicnq.com.au/osgood-schlatter-disease/>

# Osgood-Schlatter Disease



[www.bonesandspine.com](http://www.bonesandspine.com)



Source: Skinner HB, McMahon PJ: *CURRENT Diagnosis & Treatment in Orthopedics, Fifth Edition*: [www.accesssurgery.com](http://www.accesssurgery.com)

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# Clubfoot

Hindfoot deformity-  
common

Incidence 1 in 1,000-  
Males 2X more likely

Genetic  
predisposition

Involves muscular  
and skeletal parts



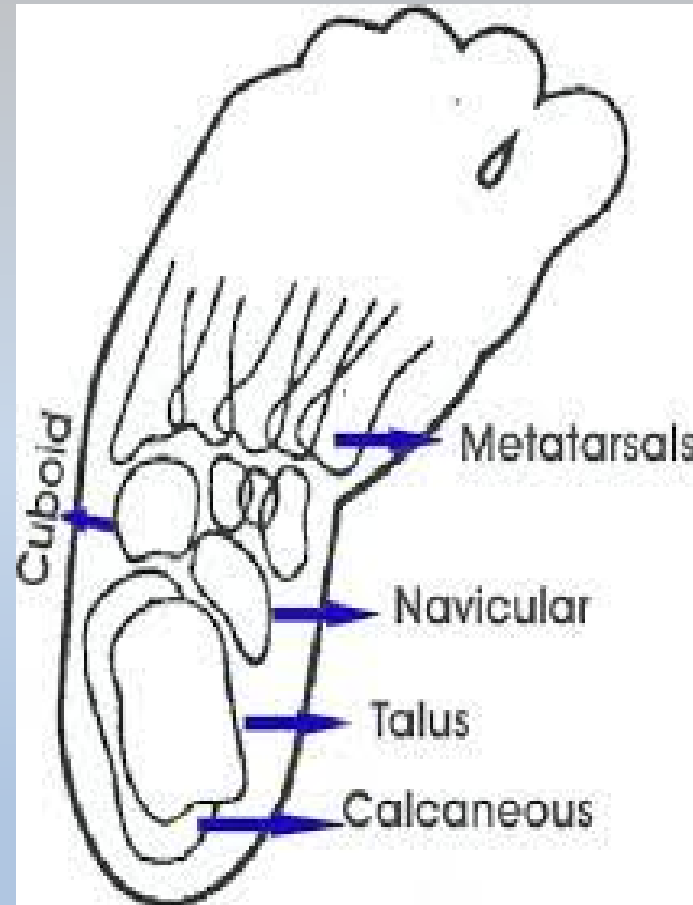
<https://www.medicalnewstoday.com/articles/183991>

# Anatomic Features

Talipes Equinovarus

Characterized by  
malalignment of the  
calcaneotalar-navicular  
complex

1. Cavus- plantar flexion  
of first ray
2. Adductus- adduction  
of the  
fore foot/midfoot on  
the hind foot
3. Varus- hindfoot points  
inward
4. Equinus- plantar  
flexion of the hind foot



[https://www.researchgate.net/figure/Bone-misalignments-in-clubfoot\\_fig3\\_309277129](https://www.researchgate.net/figure/Bone-misalignments-in-clubfoot_fig3_309277129)



# Treatment

Shortly after Birth  
Serial Casting  
Surgery may be required





# Metatarsus Adductus

Line through heel  
should cross through  
second toe

In metatarsus  
adductus, line will  
cross lateral to second  
toe.

Treatment involves  
stretching forefoot

Parents can do at  
home

Consider referral if not  
better by eight months





# Infection of Bones and Joints

Pathogenesis- how are infections acquired  
and properties unique to bone/joint

Presentation- PE and Lab findings

Bacteriology- most common causes

Management/Treatment



# Acute Osteomyelitis

## **Who gets it?**

1 in 5000 under 13 years

Most cases are in young children

Increased risk in sick children and SCD

## **Where does it occur?**

Metaphyseal end of long bones

Femur, Tibia and Humerus - 79%

## **Risk related to:**

Bones more porous

Higher rates of blood infections

Higher rates of accidents



# Pathogenesis of AHO

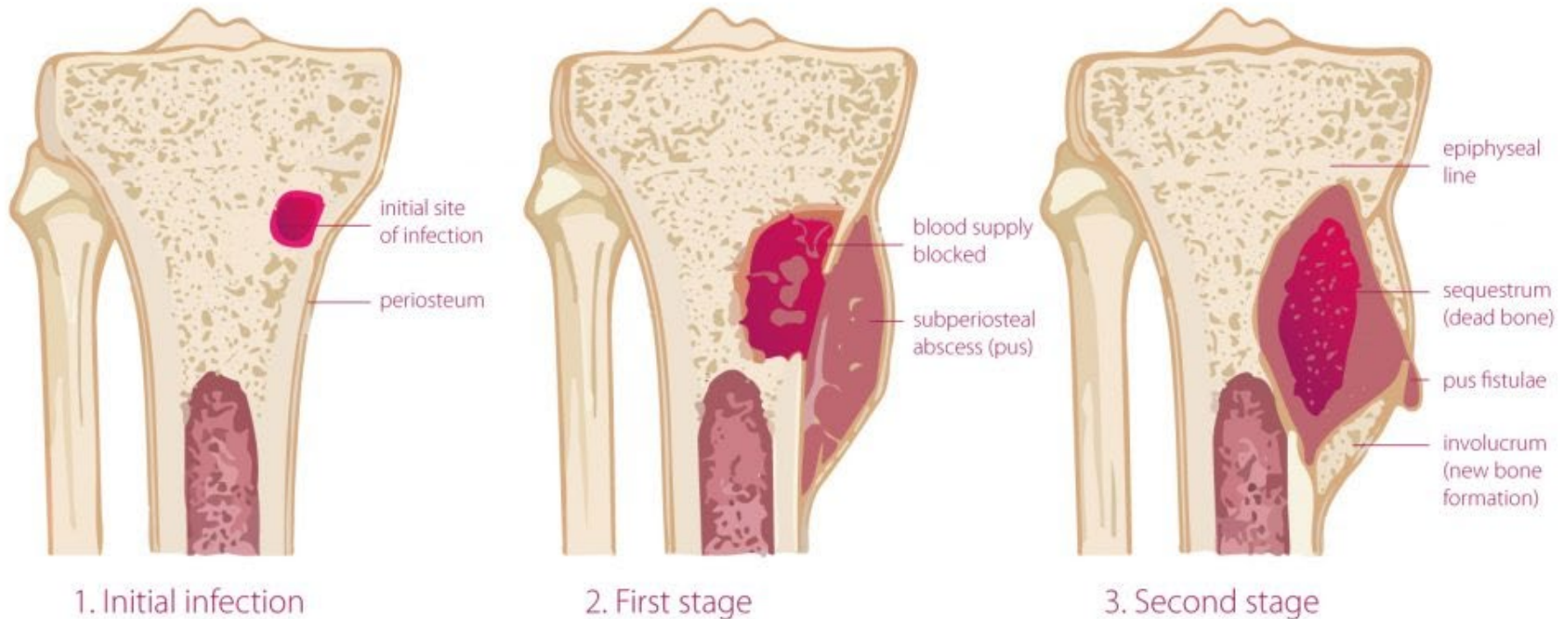
## Routes of infection

Acute hematogenous spread from serious bacterial infection – most common

Direct spread from adjacent infection- joint or sinus cavity

Direct inoculation/implantation from trauma or surgery

# Pathogenesis of AHO



development of osteomyelitis

<https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3Dfb3ewdgQ3pl&psig=AOvVaw0fLc34jRdNb15tw5zgja81&ust=1738859041248000&source=images&cd=vfe&opi=89978449&ved=0CBcQjhqxqFwoTCNCx6O74rlsDFQAAAAAdAAAAABAE>



# Bacteriology

Staph Aureus- 80% of cases over 5 y.o.

Group A Strep- a distant second – 5%

Emerging – *Kingella kingae*

Neonates- + *E.coli*/gram -, Strep sp.

Hemoglobinopathy- SCD- Gram – org.

- *Salmonella*, *E. coli* *Pseudomonas*



# History and Physical

Diagnosis is often delayed in infants

Fussiness noted first

Poor movement of extremity

Fever development

Limp if in a lower extremity

Point tenderness

Local erythema



# Diagnostic Criteria

Two of the following must be present:

Purulence of bone- biopsy or surgery

Positive blood or bone culture

Localized erythema

Positive imaging study





# Treatment

Treatment for osteomyelitis always starts with IV therapy for a minimum of 2-3 weeks

Treatment starts as “empiric”- best educated guess at bacteria then follows with culture results

Increasing MRSA has guided empiric tx to clindamycin-vancomycin- linezolid first line



# Septic Arthritis - Ditto

Sudden onset- more acute

Fever, malaise possible  
signs of sepsis

Severe pain with any  
movement of joint

Presentation will differ  
based on joint



<https://www.orthopaedia.com/septic-arthritis-2/>



# Pathogenesis

Ditto to osteomyelitis

Three different types of inoculation

Bacteriology similar

Elevated WBC and inflammatory markers

Knee > Hip > Ankle = 80% of infections

# Radiologic confirmation



<https://www.orthobullets.com/pediatrics/4032/hip-septic-arthritis--pediatric>



# Transient (toxic) Synovitis of the Hip

Common- Acute

Must be differentiated from Septic Hip

Exact etiology unclear

Generally seen ages 3-10

Preceding viral/GI infection common

Exam of hip shows pain with passive hip motion and +/- limp-

Fever – if present low grade



# TS Diagnosis and Treatment

Diagnosis is clinical

Rest

Heat

Massage

NSAIDs

Re-evaluate in 24-48 hours

Return to full activity 7-10 days



# Transient vs. Septic Hip

## Transient Synovitis

+/- Fever

Subtle onset

+/- Limp

Normal ESR

Prior illness

Normal x-ray/US

## Bacterial Septic Arthritis

High fever

Sudden onset

Limp present

ESR over 40

- Prior illness

Abnormal x-ray/US

# . Disorders of Bone Metabolism



# Rickets

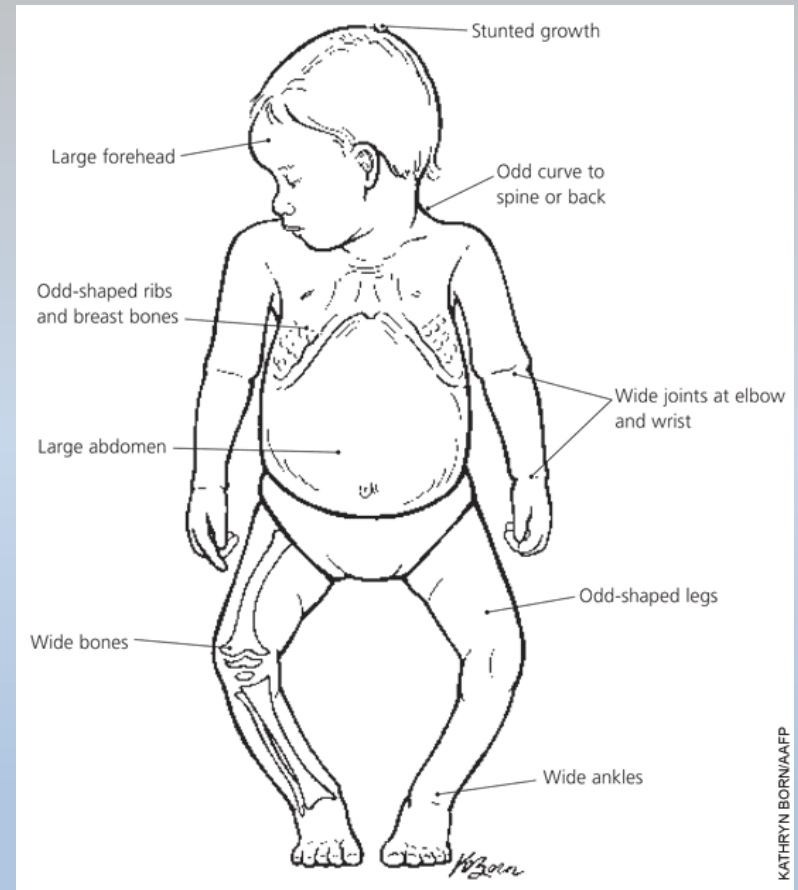
Disorder of Ca & PO<sub>4</sub> homeostasis

Severe prolonged Vit D deficiency

Bones softer & weaker

Nutritional deficiency most common

Chronic diseases may also cause



KATHRYN BORNIAFP

# Rickets

## Types of rickets

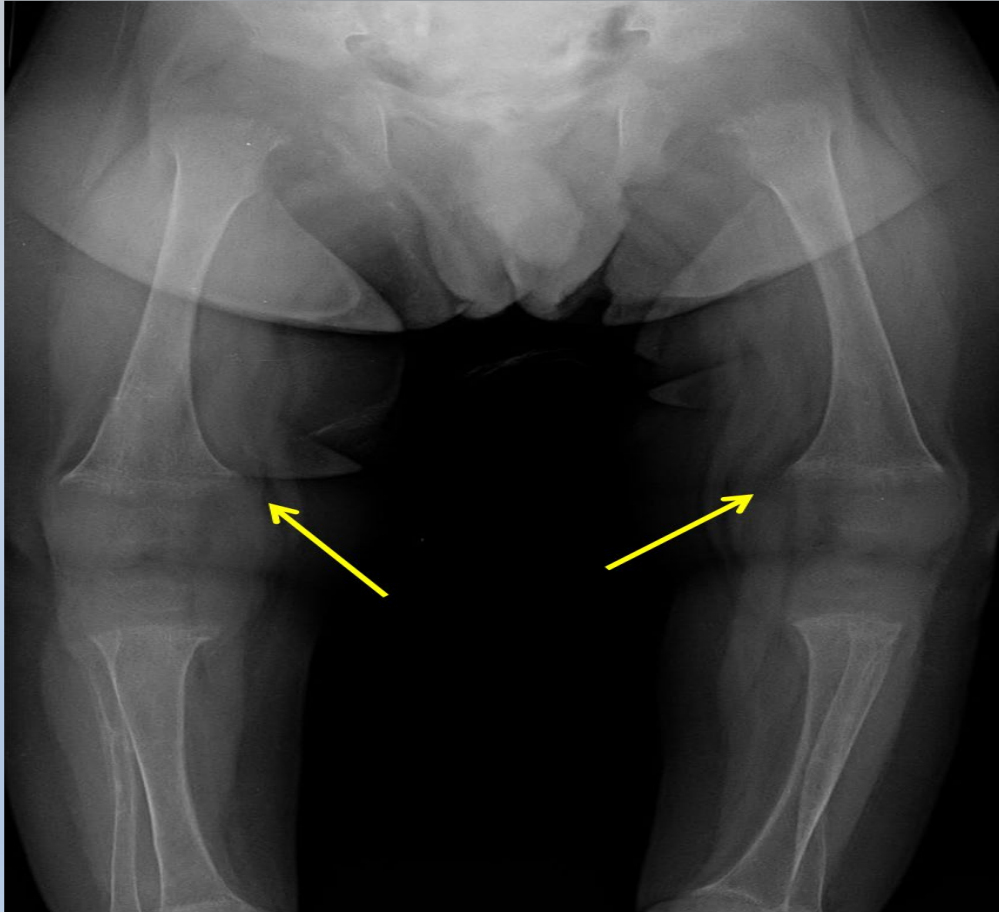
### Calcipenic rickets

- **Vitamin D deficiency or resistance**
  - Dietary deficiency
  - Malabsorption
  - Lack of sunlight exposure
  - Defect in 25 hydroxylation of vitamin D (e.g., liver disease, medications such as phenytoin)
  - Failure of 1 hydroxylation of vitamin D due to inherent deficiency of 1 alpha hydroxylase secondary to defects in the 1 alpha hydroxylase gene (VDDR I)
  - End-organ resistance to vitamin D (VDDR II)
- **Calcium deficiency**
- **Renal rickets secondary to CKD**

### Phosphopenic rickets

- **Renal tubular phosphate loss**
  - Isolated phosphate loss secondary to genetic mutations:
    - XLHR
    - ARHR
    - ADHR
    - Hypophosphatemic rickets with hypercalciuria
  - Renal Fanconi syndrome
  - Dietary phosphate deficiency
  - Phosphate malabsorption

# X-ray findings of Rickets



<https://radiologypics.com/2013/02/24/rickets-rachitic-rosary-and-differential-diagnosis/>



# Summary

1. The bones of children differ from adults and results in different types of fractures and fractures that affect the growth plate
2. Knowing the signs of accidental vs. non-accidental trauma in children is important
3. Disorders of the hip can be differentiated based on age and presentation symptoms
4. Bone and joint infections share many similarities and are both considered emergent conditions to diagnose and treat
5. Foot malformations are common in children

References are listed  
throughout this  
presentation on each  
corresponding slide

## Lecture Feedback:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

